

SHINJO, Japan

WMO: 475200

Lat: 38.7572N

Lon: 140.3120E

Elev: 102

StdP: 100.11

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.6	-5.2	-8.9	1.8	-3.4	-7.8	2.0	-2.3	10.4	-0.7	9.4	-0.9	1.8	20	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.5	32.1	24.9	30.7	24.2	29.1	23.3	25.5	30.6	24.9	29.4	24.2	28.0	3.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.1	19.2	28.0	23.5	18.5	27.3	23.0	17.9	26.7	79.0	30.9	76.3	29.6	73.6	28.1	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
9.0	7.9	6.9	-10.3	34.6	1.8	1.4	-11.6	35.6	-12.6	36.4	-13.6	37.2	-14.9	38.2		
			-10.6	26.6	1.8	0.8	-11.9	27.2	-12.9	27.6	-13.9	28.1	-15.2	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	-1.0	-0.5	2.6	8.7	15.2	19.6	23.6	24.8	20.5	13.8	6.9	1.6	(6)
(7)		DBStd	9.53	2.22	2.32	2.54	3.45	2.99	2.47	2.55	2.60	3.22	3.08	3.32	2.60	(7)
(8)		HDD10.0	1300	341	293	231	65	2	0	0	0	0	5	102	262	(8)
(9)		HDD18.3	3043	599	527	489	289	103	14	1	0	14	145	342	520	(9)
(10)		CDD10.0	1803	0	0	0	26	163	288	421	459	314	122	9	0	(10)
(11)		CDD18.3	504	0	0	0	0	6	53	163	201	78	3	0	0	(11)
(12)		CDH23.3	3344	0	0	0	5	90	283	1033	1523	402	7	0	0	(12)
(13)		CDH26.7	1013	0	0	0	0	13	51	322	538	90	0	0	0	(13)
(14)	Wind	WSAvg	2.7	3.0	3.0	3.0	3.0	2.7	2.4	2.3	2.3	2.2	2.4	2.8	(14)	
(15)	Precipitation	PrecAvg	1909	210	135	118	96	109	132	212	177	144	154	188	236	(15)
(16)		PrecMax	2196	319	220	190	155	208	284	573	450	283	240	348	440	(16)
(17)		PrecMin	1494	135	59	61	29	45	42	66	51	55	73	70	87	(17)
(18)		PrecStd	167	46	41	38	38	41	66	104	96	64	46	54	70	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	8.0	14.3	23.8	28.3	30.1	33.5	34.6	31.2	24.1	17.7	11.3	(19)
(20)			MCWB	3.6	5.0	7.8	13.3	17.7	20.7	25.2	25.6	24.2	18.5	13.6	9.4	(20)
(21)		2%	DB	4.1	6.0	11.0	20.6	25.8	28.1	31.7	32.8	29.1	22.0	15.2	8.8	(21)
(22)			MCWB	2.2	3.3	6.6	12.0	16.3	20.1	24.6	25.2	23.3	17.5	12.0	6.9	(22)
(23)		5%	DB	3.1	4.6	8.9	17.7	23.7	26.5	30.1	31.4	27.3	20.5	13.7	7.0	(23)
(24)			MCWB	1.5	2.1	5.2	11.1	15.4	19.4	23.9	24.8	22.2	16.5	11.1	5.0	(24)
(25)		10%	DB	2.1	3.3	7.1	15.4	21.5	24.8	28.4	29.8	25.7	18.9	12.2	5.5	(25)
(26)			MCWB	0.8	1.3	4.1	10.2	14.9	18.8	23.2	24.2	21.4	15.5	10.2	3.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	5.5	8.9	15.7	19.3	23.0	26.0	26.6	25.2	20.3	14.8	9.8	(27)
(28)			MCDWB	5.3	7.2	13.4	19.5	24.7	27.0	31.7	32.5	29.6	22.7	16.5	11.0	(28)
(29)		2%	WB	2.7	3.8	7.2	13.5	17.9	21.7	25.2	25.8	24.0	18.5	12.9	7.2	(29)
(30)			MCDWB	3.5	5.6	10.0	18.0	22.6	25.1	30.2	31.0	27.8	20.7	14.3	8.5	(30)
(31)		5%	WB	1.8	2.5	5.8	12.2	16.9	20.8	24.5	25.2	23.1	17.3	11.7	5.5	(31)
(32)			MCDWB	2.7	4.0	8.4	16.4	21.4	24.5	28.9	30.0	26.2	19.4	13.2	6.7	(32)
(33)		10%	WB	1.0	1.5	4.5	10.9	15.9	19.9	23.8	24.6	22.1	16.2	10.4	4.1	(33)
(34)			MCDWB	1.7	2.9	6.7	14.5	20.1	23.4	27.4	28.8	24.9	18.4	12.0	5.2	(34)
(35)	Mean Daily Temperature Range	MDBR	4.9	6.0	7.8	10.5	10.5	9.0	7.8	8.5	8.3	8.7	7.3	4.9	(35)	
(36)		5% DB	MCDBR	5.9	8.2	11.8	15.6	15.6	12.6	10.7	10.6	10.0	10.4	9.9	7.3	(36)
(37)			MCWBR	4.5	5.9	7.7	8.2	7.1	5.5	4.5	4.2	4.6	5.9	6.8	5.6	(37)
(38)		5% WB	MCDBR	5.7	7.6	10.8	12.5	11.8	9.4	9.4	9.8	8.5	8.3	8.8	7.1	(38)
(39)			MCWBR	4.6	5.9	7.6	7.5	6.2	4.6	4.2	4.1	4.2	5.3	6.7	5.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.387	0.425	0.492	0.531	0.545	0.533	0.542	0.522	0.460	0.426	0.409	0.396	(40)	
(41)		taud	1.957	1.859	1.822	1.830	1.857	1.965	2.010	2.065	2.207	2.192	2.134	2.041	(41)	
(42)		Ebn at Noon	736	763	756	758	757	766	755	758	785	767	718	695	(42)	
(43)		Edh at Noon	133	168	194	204	203	182	173	160	131	120	112	115	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.52	2.31	3.36	4.44	4.99	4.94	4.35	4.46	3.66	2.83	1.86	1.30	(44)	
(45)		RadStd	0.14	0.20	0.32	0.45	0.51	0.36	0.62	0.52	0.28	0.24	0.15	0.14	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	N/A	N/A	+0.50	N/A	+0.33	N/A	N/A	N/A	N/A	(46)
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	+0.35	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page